

Chapter 12

Application of Blockchain Technology for Distributed Management of Digital Research Library Holdings

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ABSTRACT

Technological advancement is gradually shifting the paradigm of how network records and transactions are processed and secured without third-party intervention. The importance of blockchain technology in performing these roles cannot be overstated. This technology has a wide application in the health, banking, insurance, real estate, media, and transportation sectors. Nevertheless, the role of blockchain technology in the distributed management of digital research library resources has received limited attention. This chapter discusses the distributed management of digital research library resources through blockchain technology. It explains the meaning, features, and components of blockchain technology. Concepts such as distributed library management and digital research library were clarified. The digital research library materials were highlighted along with how they can be managed. The chapter thus argues that blockchain technology can be used in the distributed management of digital research library resources.

INTRODUCTION

Blockchain technology emerged as an attempt to offer a solution to the long-lasting problem of third-party involvement in the management, exchange, and sharing of information and tenders among people

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involved in a transaction. Bitcoin has successfully applied this user-to-user data sharing or exchange model to allow free movement and exchange of digital currencies and information across different local and international locations. This end-to-end technological advancement can also be a feasible mechanism for fully implementing a distributed library management system. Although Cabello et al. (2017) proposed this model, it appears to be deficient in tackling issues of multiple library database registrations among library patrons in the global research community. Thus, this chapter was designed to explain how blockchain technology can be used to enhance the effective implementation of distributed management of research library resources and correct issues of multiple user identities in research libraries. The reader should be able to do the following after reading this chapter:

- explain the concept of blockchain technology
- state the uses of blockchain technology
- define a digital research library
- describe distributed management of research libraries
- discuss the application of blockchain in distributed management of digital research library resources

BLOCKCHAIN TECHNOLOGY

Blockchain is a highly secured and dependable transactional and information database that has received global recognition as a global record-keeping mechanism and has been found effective in managing big data (Muheidat et al., 2022). It is an integrative and decentralised database developed to record, validate, maintain, make public and distribute records of transactions among clients on the same network (Kondor et al., 2014; Sarmah, 2018; Yli-Huumo et al., 2016). It applies cryptography as a security mechanism in which every individual record, transaction, and message is cryptographically signed, thereby reducing the incidence of network hacking, data mutilation, and data compromise (Kawaguchi, 2019).

Blockchain technology is made up of blocks that allow everyone on the network to access the activities of other users in the same network. With this structure in place, it becomes more difficult for a single central block to exert influence over the network. Customers may see their transaction records without the participation of a third party, which is a vital goal of the system's architecture. The idea behind the concept of blockchain technology was to remove all barriers to the free flow of digital currency among clients in a block network across the globe through cryptography. Thus, the domain of this technological innovation originally was currency cryptography (Chen et al., 2018). Blockchain is tied to Satoshi Nakamoto, the man who formally applied the technology to operationalise the cryptocurrency called Bitcoin in 2008.

Blockchain technology is built so that no central database or block regulates, rules, or supervises it. Rawat et al. (2021) noted that to avoid the need for a trusted third party, the distributed ledger technology known as "blockchain" records all transactions and processes in an uninterrupted sequence of blocks. Every user in all the blocks in the network has adequate access to the activities of other users. Once information is verified and recorded in the blockchain, it cannot be manipulated, altered, or erased. It is sent to the network whenever a transaction is made, where automated systems verify its validity. Once transactions are validated, they are connected to the transaction log, creating what is known as a "blockchain" or "chain of transactions". The global acceptability of blockchain technology is hinged on

the level of transparency and security that characterize its operations. For instance, Kawaguchi (2019) notes that instead of all the databases on the network maintaining their separate dataset, every user has controlled access to a shared dataset. This creates a single source of truth, giving everyone working with this data confidence that they are using the most recent, accurate, and reliable dataset.

Key Operational Elements of Blockchain Technology

Blockchain technology works using three operational elements (Agrawal et al., 2021; Czachorowski et al., 2019; Kawaguchi, 2019). These are (a) a hash, (b) timestamped batches of recent valid transactions, and (c) the hash of the previous block

- **Hash:** Hash is the programme that performs the function of blockchain computation. It is a function responsible for converting data into an irregular fixed-length value. This operational element of blockchain is designed to prevent hacking as it makes it nearly impossible to guess the length of the hash, thus preventing people from hacking the network. It is a one-way function that helps scramble plaintext, data, messages and records using salt into an unreadable form that cannot be decrypted.
- **Timestamped batches of recent valid transactions:** In the blockchain network, transactions are stored in each block in a unique pattern serially. This helps to ascertain the particular or exact time the block has been mined and validated by the network. This process is called timestamping. It involves recording data on a blockchain to prove that it existed at a specific date and time. Kawaguchi (2019) notes that the timestamp element of blockchain proves that the data must have existed or been mined in the network.
- **The hash of the previous block:** In the blockchain network, as new transactions are timestamped, validated and hashed, they are linked to the hash of the previous blocks. This helps prevent the alteration of blocks or a situation where blocks are inserted between two blocks that already existed on the network. This operational element of blockchain technology is meant to perform verification functions in the network so that successive blocks strengthen the verification of the previous block. With this function, data are verified, timestamped, and recorded serially with adequate attention to time.

Features of Blockchain Technology

There are crucial distinguishing features that make blockchain technology different from other conventional digital information database systems. These include decentralised databases, corruption-proof, trackability and consensus (Andoni et al., 2019; Chen et al., 2018; Crosby et al., 2016). These are explained as follows:

- i. *Decentralisation of the database:* Distributed structures are used in the blockchain to verify data storage, management and transfer since no one block is responsible for all of the functions of the blockchain. Due to blockchain's reliance on a decentralised network of nodes, data is constantly routed via many nodes, guaranteeing that the original data's integrity is never compromised, even if one node is compromised

- ii. *Corruption-proof:* All ongoing activities among blocks in the network are accessed by all users in the blockchain structure. Also, any time a transaction is tampered with or verified data altered, all other nodes are notified using the same verification process. The chain is secured and free from corrupt practices with this operational structure. Due to the lack of a single point of failure, blockchain technology is more secure than its rivals..
- iii. *Trackability:* All transactions on the blockchain are ordered chronologically with a cryptographic hash function which enables a block to be connected to other blocks. In this way, every operation or activity is traceable using hash keys associated with a block's metadata.
- iv. *Consensus:* All blocks in the network must verify an entry or transaction for it to be authenticated or recorded on the chain. This means that all the blocks must make efforts to verify a transaction coming from each block in the blockchain. The efficiency of blockchain technology is mainly attributable to the consensus algorithm. In reality, it is a characteristic shared by all blockchains. Consensus is a method for network nodes to make collective choices. Here is where you should turn if you need a contract urgently. When millions of nodes validate a single transaction, a system's functioning requires a consensus to run smoothly. This may be contrasted with a voting system in which the majority wins, and the minority must support it. In reality, consensus destroys public faith in the system. They may not trust one another, but they may have confidence in the algorithm behind the system. Therefore, every decision taken on the blockchain is beneficial to the system.
- v. *Increased capacity:* This is the first and most important component of Blockchain. The most striking characteristic of Blockchain technology is that it increases the capacity of the whole network. Many computers working together provide greater power than a small number of centralised ones. Several computers work together, resulting in a greater overall output. This enhanced potential is shown by Stanford University's initiative to create a supercomputer capable of replicating protein folding for medical research.
- vi. *Imutability:* The establishment of immutable ledgers is one of the main advantages of blockchain. A central database is susceptible to hacking and fraud since it depends on the trust of a third-party intermediary to keep it secure. For example, Bitcoin's blockchain keeps its ledgers going forward at an unending rate. The digital ledger is available to all network nodes. Before a transaction can be added, every node must validate its authenticity. If the majority agrees, it's added to the ledger. This guarantees that the system is transparent and devoid of corruption. In addition, once transaction blocks are added to the distributed ledger, the procedure cannot be undone. As a consequence, no other network user may modify, delete, or update the file.

Components of Blockchain Technology

Blockchain technology has four essential components that enable its operations. These are (a) distributed ledger, (b) smart contracts, (c) distributed applications, and (d) the digital signature. Each of these components is discussed below.

- *Distributed Ledger:* Blockchain technology functions through distributed digital ledger technology (DLT). When using the distributed DLT, nodes (the machines that make up a distributed ledger) capture, transmit, manage, and verify transactions in each ledger.
- *Smart Contracts:* This uses a set of highly configured protocols or programs on the blockchain to automatically implement contracts and transactions between two or more parties. It ensures

transparency in the transaction(s) to increase confidence among parties without needing a third party and save time.

- *Distributed Applications:* These are programmes built into the blockchain that enable information on every transaction among users in the blocks to be recorded and shared across multiple computers on the network on a peer-to-peer basis.
- *Digital signature:* A digital signature is used to promote security in the blockchain and promote safe transactions between nodes. Two types of digital signatures are required when using blockchain technology. These include the private key (used for encrypting transactions sent to a user) and the public key (used to encrypt transactions after verification that the transaction is complete).

Uses of Blockchain Technology

The following are some uses of blockchain technology:

- Provision of peer-to-peer digital currency system:* Blockchain technology enables direct transactional interaction between blocks on the network without any intermediary. Once transactions are broadcasted on the network and verified by recipients, payment is made through a cryptographic process.
- Protection and preservation of digital records of individual blocks:* Blockchain technology prevents all verified records on each block in the network from alteration and removal. Once all the blocks in the network verify records, data, information or transaction, they are distributed and stored in the database of each block. This makes it difficult for any block to tamper with it.
- Distribution of digital records:* Another practical use of Blockchain technology is the distribution or sharing of records among the blocks in the network. The technology allows every database, in the form of a block, to be linked to another database through nodes. Thus, other blocks can assess every individual's digital records or activities in each block on the network.
- Data mining:* In blockchain technology, this is called blockchain mining and involves verifying each stage of the transaction and movement of cash from one block to the other. This process allows for quick verification of data.
- Asset tracking:* Blockchain technology is a method for monitoring assets and determining evidence of possession or origination (Fahmy, 2018). This function of blockchain technology enables the tracking of all activities carried out by clients in all the blocks of the blockchain network. This reduces the tendency of online property theft.
- Digital Identity:* This is another important use of blockchain technology. Every client is given a private and public digital key that is immutable and verified or known by every block within the chain network.

DISTRIBUTED LIBRARY MANAGEMENT

The distributed library is an emerging innovation in the library and information science discipline. It is a collection of library resources available for borrowing by users. The library catalogue is maintained on fragmented, linked together databases and accessible to users through the Internet (Melnik et al., 2021). Some researchers have observed that the distributed library system employs a vertical fragmentation

strategy, allowing for the vertical division and storage of information about loans, books, and members for all users (Khan & Hoque, 2010; San & Nyunt, 2010). A distributed library is based on a distributed database system, interconnected logically with a shared collection of data kept on computers at various locations. The data on the computer network is accessible to all users from different locations on the network. There are two types of architecture in a distributed library system: peer-to-peer (P2P) and Client-Server architectures (San & Nyunt, 2010).

A peer-to-peer (P2P) network is a distributed system of computers (known as nodes) that act as clients and servers simultaneously or separately. Under this network architecture, each node is an equal peer. Furthermore, all distributed library nodes are linked to the server hosting the network in a client-server design. Similarly, distributed library management follows the Distributed Database Management System (DDBMS) works. The DDBMS is a software that allows the administration of a distributed database and makes the dispersion of the database visible to users (Rahimi & Haug, 2010; San & Nyunt, 2010).

Each fragment of the DDBMS is kept on one or more computers under the supervision of an independent Distributed Database System (DDS) linked through communication networks to the rest of the DDBMS. Distributed Library Management (DLM) connects different distributed libraries on a single network that allows all users to have adequate access to the resources held by the various distributed libraries. According to San and Nyunt (2010), the use of the “Distributed Library Management System” (DLMS) provides access to sites from remote locations and assists in query and data transfers between different databases through the network. In addition to delivering metadata of specific distributed data storage, the DLMS also offers security control through the appropriation of access to distributed data. The DLMS also guarantees the consistency and robustness of requested data.

DIGITAL RESEARCH LIBRARY

The purpose of research libraries is to serve the research requirements of academics worldwide. While all university libraries are considered research libraries, the government and other organisations may also have specialised research libraries. A research library is often associated with scholarly communication services, such as assistance for institutionally sponsored open access publications and maintaining an institutional repository (Newton et al., 2010). Most research libraries are institutionally oriented and are intended to support the institution’s research operations. A digital research library is hosted virtually by an institution to gather, administer, preserve much digital information for the long term and provide specialised services to users and communities according to standards (Trivedi, 2010).

Collecting digital resources for information management tools does not constitute a research library. Several activities must connect library resources (collections) to people and services (Suna & Yuanb, 2012). This can be done through a complete research data cycle, including creation, dissemination, use, storage, and retrieval (Odigwe et al., 2020; Owan & Bassey, 2019). The audience of a digital research library is the worldwide community. A typical digital research library consists of four essential components: people (researchers), online information resources (such as electronic reference sources, indexes, library catalogue electronic texts and journals), and technology such as computers, the Internet, meta databases, display technologies, search, retrieval, and routing software, among others (Bansode & Shinde, 2019).

Materials in Digital Research Libraries

There are numerous materials in digital libraries that boost their functionality. These include data storage, data sharing, user interface, and data mining materials.

- i. *Data storage materials:* These are digital devices that enable the library to store and preserve information, records and transactions permanently for future use. These materials include a database and repository).
- ii. *Data sharing materials:* These are machines that enable digital libraries to move information, records, transaction, and other resources from one location to another. These include intranet and Internet.
- iii. *User interface materials:* These applications help library users interact with library resources anytime and anywhere. These include Chrome, Firefox, Edge, Brave, Opera, Internet Explorer, Safari etc. Researchers also have an emerging concern to incorporate social media tools into libraries for a more accessible user interface (Cho, 2013; Paul, 2014).
- iv. *Data mining materials:* These software applications enable digital libraries to locate, collect, prepare and store metadata about library users. Examples are web crawlers, Boolean operators and wildcards.

Managing Digital Research Library Resources

One of the problems of digital research libraries is the effective management of digital resources in their possession. This involves ensuring that users' records are not altered, research data are not tampered with, and uninterrupted access to desired resources on demand. Many researchers believe that applying Distributed Library Management System (DLMS) is a feasible measure to ensure effective management of digital research library resources (Bansode & Shinde, 2019; Chiu et al., 2021; San & Nyunt, 2010). These will involve the following practices:

- i. *Application of Distributed Database system:* Distributed databases are "logically connected" collections of shared data on a computer network that may be accessed from any system in the network (San & Nyunt, 2010). This organised approach makes digital research library materials accessible from anywhere on the network.
- ii. *Application of Peer-to-Peer (P2P) distributed system:* The application of peer-to-peer distributed database architecture can allow users to access all the activities going on in the library. This P2P distributed system can also help control illegal research practices if users or the database do not have access to the activities carried out by other users or databases.
- iii. *Linking all resources in the library to a database:* This involves clustering library holdings into fragmented databases and connecting them to a network through nodes. With this structure, every resource in the library is assigned a Digital Object Identity based on the database it belongs and thus makes accessibility easy.
- iv. *Developing a digital identity for every library user:* Digital research library resources cannot be managed when there is no model to identify those using the resources and those supplying the library. When a digital identity is used, it helps establish a link between an individual and the digital entity with which they are linked (Al-Khouri, 2014; Davis, 2014). Assigning every library user a

unique digital identity can help trace resource use to different users and track all their activities in the library.

Digital Research Libraries Initiatives

The development of Digital Research Libraries (DRL) can be traced back to the works of scientists like Vannevar Bush and J.C.R. Licklider. Their motivation and interest were to identify and develop innovative technologies and approaches toward knowledge sharing as fundamental instruments of progress (Candela et al., 2011). Digital Library Initiative (DLI) formally started in the U.S. in 1994 (Guion, 2021). This initiative was preceded by a system called “e-print archive” developed by Paul Ginsparg in August 1991 and named arXiv. It was a free distribution service for open-access scholarly writings in specific scientific disciplines. Following the operation of arXiv, the DLI emerged, which funded projects aimed at offering more of the functions of the traditional libraries in adding to collection development. Guion (2021) notes that these collections included scholarly papers and digitized maps, photographs, satellite images, videos, and more through DLI. It was possible to pull together research in digital library projects previously fragmented among discipline-specific communities. Berry (1996) notes that during this time, six U.S. institutions received funding for Digital Library Initiatives from the National Science Foundation (NSF), the Department of Defense Advanced Research Projects Agency (ARPA), and the National Aeronautics and Space Administration (NASA) beginning in the fall of 1994. These institutions were Carnegie Mellon University (Pittsburgh), The University of California at Berkeley and Santa Barbara, Stanford University (Palo Alto, CA), the University of Illinois at Urbana Champaign, and the University of Michigan (Ann Arbor).

The first-generation digital library projects were launched, including California Environmental Digital Library, Alexandra Digital Library, Informedia Digital Video Library, Interspace, and the University of Michigan Digital Library. At this time, digital research Libraries served specific communities, organizations and institutions with limited coverage. It has been argued that most of these first-generation digital libraries were “start from scratch” and had monolithic applications, lacking reusability, ease of installation, customization, and configuration (Candela et al., 2011). Though it was complicated to link different institutional digital libraries together, institutions were able to develop their digital library system. Besides, different achieves such as Electronic Thesis and Dissertation Repositories (ETDs), Achieves of Cognitive Science papers (CogPrints), and Research Papers in Economics (RePEc) were developed during this time. The inadequacy of first-generation digital libraries infrastructure to enable interoperability of institutional digital libraries reshaped the thinking of James R. Davis and Carl Lagoze about digital library infrastructure that could facilitate collaboration and sharing of research information among institutions. Two crucial research library initiatives are discussed in this chapter to offer more clarifications on the subject. These are the national and international research library initiatives, discusses hereunder.

National Digital Research Library Initiatives

In 1995, James R. Davis and Carl Lagoze of the Department of Computer Science at Cornell University developed the Networked Computer Science Technical Research Library (NCSTRL) (Davis & Lagoze, 2000). This was the first attempt to introduce the concept of open architecture in a digital library in which the library was made public, not a proprietary system owned by a particular company or community of

scholars. NCSTRL operated through Dienst, a protocol and architecture for distributed digital libraries. At this time, the concern was to build a national digital library architecture in which all institutional digital libraries could be linked together to form a national online research repository. With Open Architecture (OA), institutions could create a digital library using sets of standard protocols called “Dienst”. Davis and Lagoze (2000) note that Dienst has four core functions: it specifies the operational characteristics of core digital library services; it has a structured document model; it mandates an open, extensible protocol for communicating with digital library services and accessing these documents; and provides a mechanism for definition and administration of a distributed collection. The Dienst architecture also specifies four core digital library services - user interface, repository, index and collection services. The user interface services provide a human-friendly gateway to the information obtained from other services. According to the Dienst document model, the repository services store and provide access to documents. The index services provide search capabilities, accepting a query and returning a list of document identifiers that match the query. The collection services define the digital library collection’s components, services, and documents, making it possible for user interface services to interact with them.

The development of NCSTRL gave birth to the National Digital Library Initiative (NDLI). The first project arising from this initiative was the National Digital Library Project of the Library of Congress in the United States. Corporations and foundations are privately funded to make some of their large text and image collections publicly accessible through computer networks (Bearman, 2007). The success of the National Digital Library Project of the Library of Congress sparked the interest of other countries. Thus, in 2005, the National Digital Library Project (NDLP) was launched in China. Similarly, the National Digital Library of India (NDLI) project was initiated in 2015 and launched on June 19, 2018 (Bashir et al., 2019). In 2014, an Elsevier Foundation supported Ethiopia’s National Digital Library Project through a grant. This was officially launched in Addis Ababa, Ethiopia, on 13-15 June 2014 (The Elsevier Foundation, 2014). Pakistan’s National Digital Library Programme (NDLP) was also introduced in 2003 as part of the Higher Education Commission’s (HEC) mission to build and strengthen the research culture in Pakistan (Arshad & Ameen, 2017). However, with the development in networking worldwide, the need to design a global digital library that can enable research information sharing and data interaction across nationals. This led to the International Digital Library Initiative (IDLI).

International Digital Research Library Initiatives

The international research library initiative has been in progress since the 1970s (Berry, 1996). The Digital Library Federation (DLF) is one of the significant International research library projects designed to pioneer the use of electronic information technologies to extend collections and services. It was launched in 1995 as a program by the Council on Library and Information Resources (CLIR) (Wani, 2022). The Google Books Library Project is an international digital library project launched in 2004. In December 2004, Google announced its partnership with libraries of the University of Michigan, Harvard, Stanford, Oxford, and the New York Public Library to scan their books and make the full text searchable online. Another international digital library initiative is the Networked Digital Library of Theses and Dissertations (NDLTD). Wani (2022) notes that NDLTD is an international organization dedicated to promoting the adoption, creation, use, dissemination, and preservation of electronic theses and dissertations (ETDs). In 2009, The World Digital Library (WDL) was operated by UNESCO and the Library of Congress, U.S.A at UNESCO headquarters in Paris on April 21 (Zhang, 2011). WDL makes available on the Internet, free of charge and in multilingual format, significant primary materials from countries

and cultures worldwide. The “International children digital library (ICDL)” was designed in 2002 for children aged 3 to 13 years. The library contains digital materials collected in many languages across different countries. The ICDL was developed by the College of Information Studies and the Human-Computer Interaction Laboratory at the University of Maryland, College Park.

BLOCKCHAIN TECHNOLOGY AND THE DISTRIBUTED MANAGEMENT OF DIGITAL RESEARCH LIBRARY RESOURCES

There are several applications of Blockchain technology in the world today. For instance, Blockchain technology is used for digitising the taxation system, power loss prevention system, and health records system in Pakistan (Khalid et al., 2020; Shahnaz et al., 2019; Tan et al., 2021; Vistro et al., 2021). Blockchain technology was proposed as an application for Open Educational Resources jointly called OpenEduChain and the healthcare system in the USA (Benniche, 2019; Holotescu, 2018). Blockchain technology was proposed as an efficient blockchain-based inner-campus book sharing system called BookChian in China (Zeng et al., 2019). Although there is yet a digital library fully implementing blockchain technology worldwide, this chapter proposes its use in the distributed management of digital research libraries due to its functionality in other applications. The need for distributed management of digital library resources is very apt given the growing demand for problem-solving research and the need to make research data readily available for scholars. Apart from these, a growing concern is ensuring the integrity of data available in the global research community. Similarly, with unethical research practices on the Internet, adopting distributed management of digital library resources is very necessary. Since the distributed management of digital research library resources thrives on the web, it is imperative to implement a technology that will make the system free from all forms of unethical and corrupt research practices. Blockchain technology appears to be a promising model that corrects all anomalies associated with the operations of digital research libraries. The link between the application of blockchain technology and the functionality of distributed management of research library resources is explained using the following:

Peer to Peer Distributed Database System

Applying a peer-to-peer distributed database system, a key feature of distributed research library management needs Distributed Applications (DAs). DAs are programmes built into the blockchain to enable information on every transaction or activity among users in the blocks to be recorded and shared across multiple computers on a peer-to-peer network basis. The idea is that every digital research library is a block (distributed database) linked through nodes to form a blockchain of digital research libraries. This system will allow library users adequate access to the activities carried out by other users in any database that is connected to the chain network. Thereby reducing the chances of unethical and corrupt research practices in the libraries.

Adoption of Digital Signature (DS)

Utilising digital signatures to ensure communications security inside the blockchain network is necessary. Every user has a public and a private key (digital signature) in the blockchain. The argument here is that

when every digital library user has a Digital Signature (DS), it makes for easy tracking of all activities carried out by every digital research library user across the globe.

The Need for Distributed Ledger (DL)

Distributed management of digital library resources cannot function without distributed digital ledger technology embedded in blockchain technology. Using the Distributed digital ledger technology (DLT), transactions are recorded, shared, maintained, and synchronised amongst nodes (computers) in a distributed ledger system. Every digital research library is an independent database that records, shares, and supports research data and other library resources. The separate databases (nodes) are linked together to form a chain of digital libraries.

Distribution of Research Records

A key aspect of distributed management of digital libraries is the distribution of research records and other library resources among the distributed databases within the network. Blockchain applies this system to share transactional records among the blocks in the network. The application of this system in the distributed management of digital libraries will allow every digital library to access research data, information and activities of other library users in the network. This makes library resources and records immutable and secured.

Data Mining

Distributed management of the digital library involves the movement of records, information, and research data from one digital library to another within the library chain. This approach is called blockchain mining and involves verifying each stage of the transaction and movement of cash from one block to the other. This application will allow for adequate verification of library users' research records and other library resources moving from one digital library to another.

CONCLUSION

The distributed management of digital libraries is an innovation in library and information practice that gives a new dimension to managing resources effectively. Different researchers have recommended the application of distributed management of digital libraries as a feasible model for managing digital research library resources within an institution. However, there appears to be limited knowledge on how it can be applied in the global digital research library network. Besides, there is little information on how users' records, research data and other library resources can be distributed across different digital research libraries and made accessible to users with a high level of integrity. As discussed in this chapter, blockchain technology promises to be a pragmatic model that can drive distributed management of digital library resources at the global level. This is because it has all the applications that can effectively distribute digital research library resources among digital research libraries (distributed databases) and reduce the chances of unethical and corrupt practices within the global research community.

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KEY TERMS AND DEFINITIONS

Blockchain: A blockchain is a network system that execute transactions among connected nodes using high level security.

Data Mining: Data mining is a process of extracting and verifying data or information on a network.

Database System: Database system is a warehouse that manages information.

Digital Research Library: This is an online database that hosts diverse researcher materials that can be inter-operated or remotely accessed by different users on the internet.

Library User: This is any individual that uses the library.

Management: Management is the process of planning, organizing, directing, coordinating and controlling human and material resource to achieve a goal.

Network: A network is a collection of computers that allow for data sharing.

Application of Blockchain Technology for Distributed Management of Digital Research Library Holdings

Peer-to-Peer System: This a network system involving two nodes without a server.

Research Records: These are source documents or files for verification of research information on a network.

Technology: This refers to the application of scientific knowledge to solve practical problems.